

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

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B4: NUCLEAR AND PARTICLE PHYSICS

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TRINITY TERM 2025

Originally set Friday, 13 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

*The use of approved calculators is permitted.*

*A list of physical constants and conversion factors accompanies this paper.*

*The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.*

**Do NOT turn over until told that you may do so.**

1. (a) Define the differential scattering cross-section per solid angle  $\frac{d\sigma}{d\Omega}$  in terms of measurable quantities, for the case of a beam of particles of type  $X$ , with areal flux density  $j_X$ , scattering elastically off a single stationary target particle of type  $Y$ . State the dimensions of all properties involved.

Reformulate this definition for the case of a solid target with an incoming beam, of  $X$ -particles of current  $I_X$ . The target has area  $A$  perpendicular to the beam, thickness  $\Delta x$  parallel to the beam and number density  $\rho$ .

How can you measure the differential scattering cross-section and what requirements are there to the geometry of the target?

Define the total scattering cross section  $\sigma$ . State how  $\sigma$  relates to the thickness of the target  $\Delta x$  and the change in beam current  $\Delta I_X$  after passing through the target. [7]

(b) Draw all first order Feynman diagrams by which neutrinos of any flavour can interact with the leptons in ordinary matter. [2]

(c) Assume the total flux  $\Phi_\nu$  of neutrinos of all flavours to be  $1 \times 10^{11} \text{ cm}^{-2}\text{s}^{-1}$ . Let the total scattering cross-section for the reaction  $\nu + e^- \rightarrow \nu + e^-$  averaged over all neutrino flavours be  $\sigma_{\nu+e \rightarrow \nu+e} = 10^{-45} \text{ cm}^2$ . In a detector using  $17,400 \times 10^3 \text{ kg}$  of an unspecified detection medium, on average 5 neutrinos interact with electrons per day. Estimate how many electrons are in the detection medium. From this, deduce whether the detection medium is  ${}^4_2\text{He}$ ,  ${}^{72}_{32}\text{Ge}$ , or water (oxygen is  ${}^{16}_8\text{O}$ ). [5]

(d) All neutrinos interact with leptons in ordinary matter through a neutral intermediate bosons. How can these bosons be experimentally produced ‘on shell’ using lepton beams? Draw the relevant Feynman diagram for this production process. [2]

(e) Write down the form of the production cross-section for  $\nu_e \bar{\nu}_e$  pairs in unpolarised  $e^+e^-$  collisions as a function of the spins of the particles and the CoM energy and momentum near 90 GeV. Define the meaning of all symbols in your expression. At which CoM energy will the production rate of  $\nu \bar{\nu}$  pairs be maximal? [5]

(f) Outline how measurements of the cross-section of  $\sigma(e^+e^- \rightarrow \mu^+\mu^-)$  and  $\sigma(e^+e^- \rightarrow \text{hadrons})$  versus CM energy, and knowledge of the partial width of the Z-boson into electron neutrinos  $\Gamma_{\nu_e \bar{\nu}_e}$ , can be used to infer the number of light neutrino generations. [4]

2. (a) Use the SEMF below to estimate the change in binding energy when a thermal neutron enters  ${}^{235}_{92}\text{U}$  and  ${}^{238}_{92}\text{U}$ . Which term in the SEMF primarily causes the difference between these two changes?

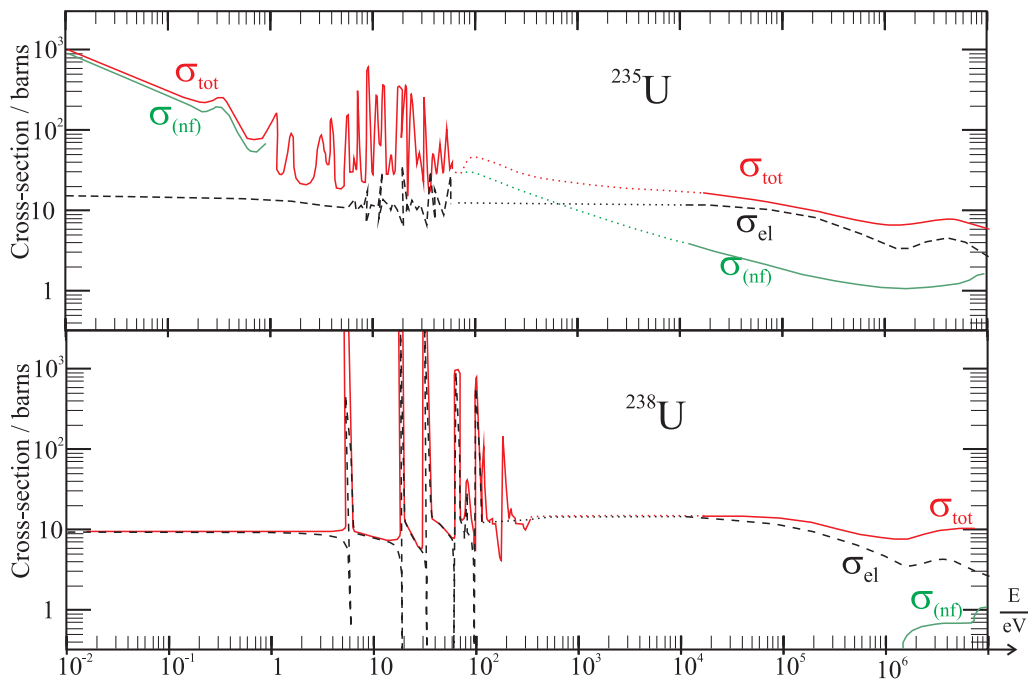
$$M(A, Z)c^2 = M_p c^2 Z + M_n c^2 (A - Z) - \alpha A + \beta A^{2/3} + \epsilon Z^2 A^{-1/3} + \gamma \frac{(A - 2Z)^2}{A} + \delta.$$

In MeV:  $\alpha = 15.56$ ,  $\beta = 17.23$ ,  $\epsilon = 0.697$ ,  $\gamma = 23.285$ ,  $\delta = \pm 12A^{-1/2}$  or 0. [6]

(b) Sketch the potential energy of the fission fragments of a heavy nucleus versus their separation. Indicate the activation energy  $E_f$  and  $Q$ -value in your sketch. Explain qualitatively how the shape arises. Why is there a minimum energy  $E_{\min}$  a neutron has to bring to the nucleus to induce fission? How can spontaneous fission occur? [7]

(c) The figure below shows the total  $\sigma_{\text{tot}}$ , elastic  $\sigma_{\text{el}}$  and induced fission cross-sections  $\sigma_{\text{nf}}$  for  ${}^{235}_{92}\text{U}$  and  ${}^{238}_{92}\text{U}$  versus neutron kinetic energy  $E$ . Dotted lines indicate average cross-sections where narrow resonance cannot be resolved. Given that the activation energies for  ${}^{235}\text{U}$  ( ${}^{238}\text{U}$ ) are 5.0 MeV (5.9 MeV), explain why  ${}^{238}\text{U}$  has negligible  $\sigma_{\text{nf}}$  for  $E < 1$  MeV but  ${}^{235}\text{U}$  has significant  $\sigma_{(\text{n},\text{f})}$  for  $E > 0$  MeV.

List the typical products of uranium fission. How can fission reactors be controlled on a timescale of seconds? Why can fission reactors not be switched off and left alone? [6]



(d) Let  $p = 0.12$  (0.79) be the probability that a neutron of kinetic energy  $E = 2$  MeV (0.07 eV) induces fission *before* being absorbed by other processes in natural uranium containing  $c = 0.7\%$  of  ${}^{235}_{92}\text{U}$  by number. Given that fission produces on average  $\nu = 2.5$  neutrons of  $E \approx 2$  MeV, find an expression for the neutron reproduction or multiplication factor  $\eta$  in terms of  $p$  and compute  $\eta$  for  $E = 2$  MeV (0.07 eV). Comment on the implications for a sustained chain reaction. How can a fission reactor use natural uranium as fuel? [3]

(e) Derive  $p$  from the plots and the value of  $c$ . Compute  $p$  at  $E = 2$  MeV (0.07 eV). [3]

**3.** (a) Using Fermi's Golden Rule for the decay rate per energy interval,  $\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |M_{if}|^2 \rho$ , and the definition of the differential scattering cross-section  $\frac{d\sigma}{d\Omega}$ , derive explicitly the non-relativistic formula given below,

$$\frac{d\sigma}{d\Omega} = |f(\underline{k}, \underline{k}')|^2 = \frac{m^2}{(2\pi\hbar^2)^2} \left| \int e^{-i\underline{q} \cdot \underline{r}} V(\underline{r}) d^3\underline{r} \right|^2,$$

where  $V(\underline{r})$  is the scattering potential. State clearly all assumptions and show the intermediate steps of the calculation. Define all symbols. Explicitly explain how the interaction Hamiltonian  $\hat{H}$  between the target and projectile enters the calculation. [10]

(b) Calculate explicitly the scattering amplitude  $f(\underline{k}, \underline{k}')$  for the spherical Yukawa potential  $V_{\text{Yuk}} = \alpha \frac{e^{-\mu r}}{r}$ . Which physical model of particle interactions gives rise to the Yukawa potential?

How do the parameters of the potential relate to this model?

Give an example of sub-atomic interactions that is well described by the Yukawa potential. [7]

(c) From your result in b), find  $\frac{d\sigma}{d\Omega}$  in terms of the scattering angle  $\theta$  and the kinetic energy  $T$  of the projectile. From the above result, derive the differential Rutherford scattering cross-section. Which interaction gives rise to Rutherford scattering? What is the total scattering cross-section in the Rutherford case? Which property of the interaction causes the divergent behaviour? [8]

4. A particle decays into the final state containing only  $D^- + \pi^+$ . Below we list properties of the six lightest potential mother particles and the final state particles.

|          | $m/\text{MeV}c^{-2}$ | $\tau/\text{s}$        | quarks     |             | $m/\text{MeV}c^{-2}$ | $\tau/\text{s}$        | quarks     |
|----------|----------------------|------------------------|------------|-------------|----------------------|------------------------|------------|
| $B^0$    | 5276.66              | $1.52 \times 10^{-12}$ | $d\bar{b}$ | $\Lambda_B$ | 5619.60              | $1.57 \times 10^{-12}$ | $ubd$      |
| $B^{*0}$ | 5324.71              | $2 \times 10^{-15}$    | $d\bar{b}$ | $\bar{D}^0$ | 1864.84              | $4.1 \times 10^{-13}$  | $\bar{c}u$ |
| $B_1^0$  | 5721.01              | $1.67 \times 10^{-23}$ | $d\bar{b}$ | $B^+$       | 5279.34              | $1.64 \times 10^{-12}$ | $u\bar{b}$ |
| $D^-$    | 1869.66              | $1.03 \times 10^{-12}$ | $\bar{c}d$ | $\pi^+$     | 139.57               | $2.60 \times 10^{-8}$  | $u\bar{d}$ |

(a) Explain with reasons which mother particles can in principle decay into  $D^- + \pi^+$ . [4]

(b) Draw a leading order Feynman diagram for such a decay. [2]

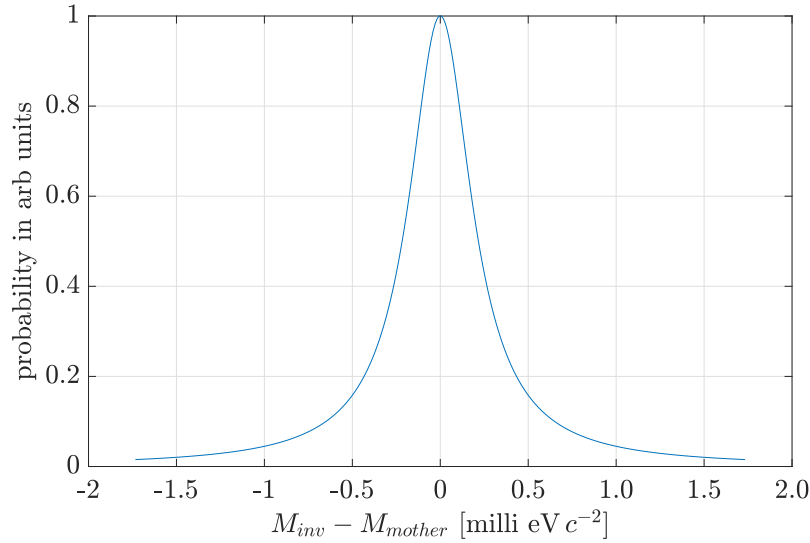
(c) In an experiment, the momenta of the decay products were measured with an uncertainty of 2%. In one event, they were found to lie in the  $x$ - $y$  plane as follows:

$$p_\pi = 2.3182 \text{ GeV}/c, \text{ direction } +0.4081\mathbf{i} + 0.9129\mathbf{j}$$

$$p_D = 2.3182 \text{ GeV}/c, \text{ direction } -0.4081\mathbf{i} - 0.9129\mathbf{j}.$$

Use the momenta to rule out mother particles. Which mother particles remain? [4]

(d) Consider an imaginary detector with perfect resolution. After many measured events the final state invariant mass minus the mass of the mother would look as follows:



Use the plot to argue which of the remaining candidates is the mother. [3]

(e) Sketch the invariant mass distribution for the decay of the candidate ruled out in part (d) and label your sketch axes. [1]

(f) Given that the  $B^{*0}$  and  $B^*$  have opposite parity and are the lightest  $d\bar{b}$  mesons, argue what the quark model predicts for  $(I, J^P)$  of  $B^0$  and  $B^*$ . [4]

(g) Given that the  $B^*$  decays to  $B^0 + X$ , how are  $B^*$  and  $B^0$  related and what is  $X$ ? Give the total energy  $X$  must have. [3]

(h) In  $e^+e^-$  collisions pairs of mother, anti-mother particles are produced. Tracks from the decay vertices of these particles originate on average  $L_0 = 0.2 \text{ mm}$  away from the beam interaction point. What is the centre of mass energy of the collider? [4]